



Software Requirements Specification

BPG Platform · Version 1.0 · Prepared for software development handoff

Document	Software Requirements Specification (SRS)
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Purpose of this document. This SRS translates the frozen BPG v1.0 design into precise, testable software requirements suitable for direct handoff to a development team. It defines scope, roles, functional and non-functional requirements, workflows, business rules, permissions, and acceptance criteria. It does not contain implementation code or visual redesign.

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1. Executive Summary

The Balochistan Procurement Guide (BPG) is an independent, AI-assisted procurement and legal-research knowledge platform. It organises publicly available Acts, Rules, Regulations, Circulars, Notifications, Court Judgments, PRC Decisions, Standard Bidding Documents, engineering standards and research into one searchable, citable repository, and layers an AI Copilot that explains official material in plain language while preserving the original legal text.

The product targets contractors, consultants, engineers, procuring agencies, lawyers, students and researchers across Pakistan, beginning with Balochistan. The v1.0 design is frozen; this SRS specifies the system to be engineered against that design. The guiding principle throughout is **official sources first — AI explains, it never replaces the law.**

2. Vision & Objectives

Vision

To become Pakistan's most trusted and comprehensive procurement, engineering and infrastructure knowledge platform through organised, searchable and AI-assisted access to official resources.

Objectives

- Provide a single, authoritative repository of public procurement knowledge.
- Reduce the time professionals spend locating and interpreting rules, decisions and judgments.
- Deliver AI explanations that are transparent, cited and clearly separated from official text.
- Enable verified community contribution to expand coverage over time.
- Maintain trust through versioning, citations, correction mechanisms and clear independence from government.

3. Project Scope

In scope (v1.0)

- Public knowledge platform: Home, Search, Repository, Document Viewer, Document details, AI Workspace, Resources, Procurement Tools, About, Contact, Contribute.
- Administration: Enterprise Admin dashboard and Contribution Review workspace.
- Authentication, roles & permissions; bookmarks, collections, downloads, notifications, settings.
- Document lifecycle (ingest → OCR → index → review → publish → version) and community contribution workflow.
- AI Copilot with reading modes, citations and confidence; full-text and semantic search.

Out of scope (v1.0)

- Native mobile applications (responsive web only; responsive layouts are the first build-phase task).
- Paid subscriptions / billing; third-party API marketplace; offline desktop apps.
- Dark mode (deferred to v1.1); international (World Bank/ADB/IsDB) libraries (v1.1+).

4. Business Goals

- **Adoption:** become the default reference for procurement professionals in Balochistan within 12 months of launch.
 - **Coverage:** grow from the launch corpus to broad federal + provincial coverage via verified contributions.
 - **Trust:** zero published documents without source verification; every AI answer carries a citation.
 - **Efficiency:** measurable reduction in research time (target: a relevant document or answer in under 30 seconds).
 - **Sustainability:** a maintainable platform and contribution pipeline that scales nationally.
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5. Stakeholders

Stakeholder	Interest
Product Owner / Founder	Vision, roadmap, content integrity, final approval.
End users (contractors, engineers, consultants, lawyers, officers, students, researchers)	Fast, accurate, citable access to procurement knowledge.
Contributors	Submit documents; receive status and recognition.
Moderators / Reviewers	Verify and publish contributions accurately and quickly.
Administrators	Operate the repository, manage users, monitor health.
Development team	Clear, testable requirements and acceptance criteria.
Issuing authorities (BPPRA, PPRAs, courts, PEC, NHA...)	Accurate attribution; correction channel.

6. User Roles

Role	Description & core capabilities
Guest	Unauthenticated visitor. Browse, search, read documents, use AI in limited/anonymous mode, use tools, view resources, submit a contribution.
Registered User	Authenticated. All guest abilities plus bookmarks, collections, saved AI conversations, downloads, history, notifications, settings.
Contributor	Registered user with a tracked contribution history and recognition; may submit and track documents through review.
Moderator	Reviews the contribution queue: OCR, metadata, duplicate & source verification; approve, reject, merge, archive, request info, publish.
Administrator	Full repository & content management, user management, categories, authorities, analytics, repository health, AI/OCR queues.
Super Administrator	All admin abilities plus role assignment, system settings, backups, version control, audit configuration and destructive operations.

7. Functional Requirements

Requirements are identified as `FR-{module}-{n}` . Each is independently testable.

7.1 Homepage

- `FR-HOME-1` Display hero, global search entry, popular queries, featured collections, latest documents and trust/disclaimer messaging.
- `FR-HOME-2` Global search input shall route the query to the Search results screen, preserving the query string.
- `FR-HOME-3` All navigation, quick actions and featured cards shall route to their corresponding modules.

7.2 Search

- `FR-SEARCH-1` Provide keyword full-text search across all published documents and metadata.
- `FR-SEARCH-2` Provide semantic (meaning-based) search and OCR-text search over scanned documents.

- **FR-SEARCH-3** Provide faceted filters (type, authority, province, year, status) and sorting (relevance, newest, oldest).
- **FR-SEARCH-4** Provide autocomplete suggestions, recent searches and popular searches.
- **FR-SEARCH-5** Present an optional AI answer card with cited sources above ranked results.
- **FR-SEARCH-6** Provide a no-results state offering filter reset and an "ask AI" fallback.
- **FR-SEARCH-7** Paginate results; maintain filter/sort state across pages.

7.3 Repository Browser

- **FR-REPO-1** Browse by collection (legal framework, PRC, judgments, circulars, research, bidding documents, engineering, authorities).
- **FR-REPO-2** Faceted filtering by type and authority; sorting by newest, oldest, title.
- **FR-REPO-3** Toggle grid and list views; display total counts and repository statistics.
- **FR-REPO-4** Each document card shall show type, authority, year, status and snippet, and link to the document.

7.4 Document Viewer

- **FR-DOC-1** Render documents in Reading, Text and Pages views with zoom, rotate and focus (fullscreen) mode.
- **FR-DOC-2** Provide a contents outline with scroll-synced highlighting and click-to-jump.
- **FR-DOC-3** Provide find-in-document with highlighted matches, match navigation and counts.
- **FR-DOC-4** Provide an intelligence panel: AI summary/actions, metadata, related content, timeline, references.
- **FR-DOC-5** Provide actions: bookmark (persisted), copy citation, open official source, download, print, share.
- **FR-DOC-6** Document status (In force / Amended / Repealed / Superseded / Draft / Historical) shall display as colour + label.

7.5 AI Workspace (Copilot)

- **FR-AI-1** Provide a conversational interface grounded in the official corpus; answers must cite sources.
- **FR-AI-2** Each answer shall support reading modes: **Official** (verbatim text), Professional, Plain English, Legal Analysis, Procurement Practice. Official text must remain separate from AI explanation.
- **FR-AI-3** Each answer shall present key points, practical application, cross-references, citation and a confidence level (High / Medium / Limited authority).

- **FR-AI-4** Provide quick actions (explain, summarize, compare, generate note/clarification/complaint/PRC appeal, generate citation) and auto-generated follow-up suggestions.
- **FR-AI-5** Provide a context panel of official sources, related documents and timeline for the active answer.
- **FR-AI-6** Every answer shall display the disclaimer that the official published document remains authoritative.
- **FR-AI-7** Registered users may save, revisit and export conversations (PDF, Word, copy, print, share).

7.6 Resources

- **FR-RES-1** Present categorised official external links (procurement authorities, engineering bodies, finance institutions, international partners) opening in new tabs with external-link affordance.

7.7 Procurement Tools

- **FR-T00L-1** Provide working calculators: Bid Capacity, Performance Guarantee, Bid Security, Working Capital, JV Share, CSR, Price Adjustment, Liquidated Damages, Time Extension.
- **FR-T00L-2** Each calculator shall compute live on input, show the formula, and present indicative-only disclaimers.

7.8 Community Contribution

- **FR-CONTRIB-1** Provide drag-and-drop and browse upload accepting PDF, DOCX and ZIP (≤50 MB/file).
- **FR-CONTRIB-2** Capture metadata: title, authority, type, province, category, document number, year, language, source URL, source type, description.
- **FR-CONTRIB-3** Support optional contributor identity and anonymous submission; require a declaration of public availability and accuracy.
- **FR-CONTRIB-4** Validate inputs inline (required title, valid email, file present, declaration accepted) with field-level errors.
- **FR-CONTRIB-5** On submission, generate a unique reference number and display a submission-status tracker.
- **FR-CONTRIB-6** Allow contributors to track status: Pending, Under Review, Approved, Rejected, Need More Information, Duplicate.

7.9 Contribution Review (Moderation)

- **FR-REVIEW-1** Present a filterable submission queue with status and metadata.
- **FR-REVIEW-2** Provide document preview, submitted metadata, OCR-extracted text and AI summary/metadata suggestions.
- **FR-REVIEW-3** Provide duplicate detection with a similarity score and links to candidate matches.
- **FR-REVIEW-4** Provide official-source verification and cross-reference review.
- **FR-REVIEW-5** Provide decision actions: Approve, Reject, Merge, Archive, Publish, Request Information, Edit Metadata.
- **FR-REVIEW-6** Record a full history/audit trail per submission and present the processing workflow state.

7.10 Admin Dashboard

- **FR-ADMIN-1** Present operational KPIs: pending reviews, repository health, pending AI jobs, duplicate alerts, missing documents, active users, downloads, searches, storage.
- **FR-ADMIN-2** Present analytics: repository growth, documents by authority, downloads, search trends, AI usage, community contributions.
- **FR-ADMIN-3** Provide management modules: documents, categories, authorities, content types, engineering libraries, users, roles.
- **FR-ADMIN-4** Provide system modules: AI, repository health, search index, OCR queue, logs, backups, version control, settings.
- **FR-ADMIN-5** Surface a recent-activity feed and route to the review queue.

7.11 Authentication

- **FR-AUTH-1** Support email/password sign-up and sign-in, email verification and password reset.
- **FR-AUTH-2** Support optional organisation SSO and secure session management with expiry and refresh.
- **FR-AUTH-3** Enforce role-based access on every protected route and action (see Permissions Matrix).

7.12 Notifications

- **FR-NOTIF-1** Notify users of submission status changes, replies to "request information", and published documents they follow.
- **FR-NOTIF-2** Notify moderators/admins of new submissions, duplicate alerts and system events.
- **FR-NOTIF-3** Provide in-app notification centre and optional email delivery with per-type preferences.

7.13 Bookmarks & 7.14 Collections

- **FR-BMK-1** Registered users may bookmark documents and clauses; bookmarks persist server-side.
- **FR-COLL-1** Users may organise bookmarks and documents into named collections and share them (optional, read-only links).

7.15 Downloads

- **FR-DL-1** Provide document downloads (PDF/DOCX where available) and AI-answer/document export to PDF and Word; log each download.

7.16 Analytics & 7.17 Settings

- **FR-ANALYTICS-1** Capture privacy-respecting usage analytics (searches, views, downloads, AI usage, coverage) for admin reporting.
- **FR-SETTINGS-1** Users may manage account, notification preferences, language (EN/UR), privacy and connected apps.

8. Non-Functional Requirements

Attribute	Requirement
Performance	P95 page interactive ≤ 2.5s on broadband; search results ≤ 1s for keyword, ≤ 3s for semantic; AI first token ≤ 3s.
Availability	≥ 99.5% monthly uptime; graceful degradation if AI/search services are unavailable.
Security	Encryption in transit (TLS 1.2+) and at rest; OWASP Top-10 mitigations; RBAC enforced server-side.
Scalability	Support 100k+ documents and 10k concurrent users via stateless services and horizontal scaling; list virtualisation.
Accessibility	WCAG 2.1 AA: AA contrast for body text, keyboard navigation, focus states, ARIA labels, status by colour + text.
Maintainability	Component-driven UI, documented APIs, automated tests, linting and CI.
Reliability	Idempotent ingestion; retry/queue for OCR/AI jobs; no data loss on failure.
Usability	Any document or answer reachable in ≤ 3 clicks; consistent components; clear empty/error/success states.
Responsive	Functional across mobile, tablet, laptop, desktop and ultra-wide; touch targets ≥ 44px.

9. User Stories

As a contractor, **I want** to find the exact rule on bid security and read the official text, **so that** I can comply confidently before bidding.

As a consultant, **I want** the AI to compare two rules with citations, **so that** I can advise a client quickly and defensibly.

As an engineer, **I want** calculators for performance guarantee and price adjustment, **so that** I can prepare figures without manual error.

As a lawyer, **I want** binding vs persuasive judgments clearly tagged with citations, **so that** I can build an argument efficiently.

As a procurement officer, **I want** the latest amendment and its history, **so that** I apply the current rule.

As a student/researcher, **I want** plain-English explanations linked to official sources, **so that** I can learn accurately.

As a contributor, **I want** to upload a document and track its review, **so that** I know when it is published.

As a moderator, **I want** OCR, duplicate detection and source verification in one place, **so that** I can approve accurately and fast.

As an administrator, **I want** repository health and analytics, **so that** I can manage coverage and quality.

As a registered user, **I want** bookmarks, collections and saved AI chats, **so that** I can resume my research across devices.

10. Use Cases

ID	Use case	Actor	Outcome
UC-01	Search and open a document	Guest	Ranked results → document opened in viewer.
UC-02	Ask the AI Copilot	Guest/User	Cited, mode-aware answer with confidence + sources.
UC-03	Bookmark & collect	Registered User	Document saved to a collection, synced.
UC-04	Submit a contribution	Contributor	Reference number issued; status tracked.
UC-05	Review & publish a contribution	Moderator	Verified document published to repository.
UC-06	Manage users & roles	Admin/Super Admin	Role assigned; permissions updated.
UC-07	Monitor repository health	Admin	KPIs, alerts and analytics reviewed.
UC-08	Calculate a procurement figure	Guest/User	Indicative result with formula.

11. Business Rules

- **BR-1** No document is published without source verification by a moderator.
- **BR-2** Official text is stored and displayed verbatim; AI explanations are stored and displayed separately and never alter official text.
- **BR-3** Every AI answer must include at least one citation to an official source and a confidence level.
- **BR-4** Superseded/repealed documents remain accessible and are clearly labelled; the current version is linked.
- **BR-5** Bid security $\leq 5\%$ and performance guarantee $\leq 10\%$ are enforced as calculator/validation guidance per BPPR 2014.
- **BR-6** Anonymous contributions are accepted but undergo the same verification; contributor identity is never published without consent.

- **BR-7** Duplicate submissions above the configured similarity threshold are flagged for merge/reject.
- **BR-8** A correction request channel exists for issuing authorities; corrections are versioned, not overwritten.

12. Permissions Matrix

Capability	Guest	User	Contrib.	Mod.	Admin	Super
Browse / search / read	✓	✓	✓	✓	✓	✓
Use AI (limited)	✓	✓	✓	✓	✓	✓
Bookmarks / collections / save chats	—	✓	✓	✓	✓	✓
Submit contribution	✓	✓	✓	✓	✓	✓
Track own contributions	ref#	✓	✓	✓	✓	✓
Review queue / OCR / duplicate	—	—	—	✓	✓	✓
Approve / reject / merge / publish	—	—	—	✓	✓	✓
Manage documents / categories / authorities	—	—	—	—	✓	✓
Manage users / analytics / health	—	—	—	—	✓	✓
Assign roles / settings / backups / version control	—	—	—	—	—	✓

13. Document Lifecycle

Ingest → OCR → Index → Metadata → Source verification → Review → Publish → Versioned → (Amend / Supersede / Repeal / Archive)

- Each document carries: type, authority, province, number, year, language, status, version, effective date, last updated, keywords, source URL and relationships (related acts/rules/circulars/notifications/judgments).
- Status transitions are role-gated and audited. Amendments create a new version linked to prior versions; nothing is hard-deleted in v1.0 (soft-archive only).

14. Community Contribution Workflow

Upload → OCR → Duplicate detection → Metadata verification → Official-source verification
→ Approval → Repository → Website

- Submission issues a reference number and Pending status; the contributor is notified on every transition.
- Moderator may Approve, Reject (with reason), Merge (into an existing document), Archive, or Request Information; "Need More Information" pauses the clock pending contributor reply.
- On publish, the document enters the repository, is indexed for search and AI, and followers are notified.

15. AI Workflow

User query → Retrieve official sources (RAG) → Compose grounded answer → Attach citations + confidence → Render in selected reading mode (Official text kept separate) → Offer follow-ups / export

- Answers must be retrieval-grounded; if confidence is Limited or no source is found, the system says so rather than fabricating.
- Official mode returns verbatim stored text; explanation modes never modify it. Disclaimer is always shown.

16. Search Workflow

Query → (keyword + semantic + OCR) retrieval → Rank → Facet/sort → Optional AI answer card → Results / empty state

- Index covers title, body, OCR text and metadata. Suggestions draw from query history and popular queries.
- Ranking blends lexical relevance, recency and authority weighting; filters narrow the candidate set before ranking.

17. Repository Workflow

Browse by collection or facet → open document detail → open viewer. Repository statistics, version history, AI summary and cross-references are available per document. List virtualisation is required for large result sets.

18. Notification Workflow

Event (status change / reply / publish / duplicate / system) → Resolve recipients (role + follow) → Deliver in-app (+ email if opted) → Mark read / preferences

19. Error Handling

- Inline, field-level validation with clear messages on all forms (contribution, contact, auth, settings).
 - Friendly empty states (no results, empty collections) and recoverable error states (failed upload/OCR/AI) with retry.
 - Global 404 routes to a helpful page; unknown/under-construction modules route to a clearly labelled placeholder.
 - Service failures (AI/search) degrade gracefully and never block reading official documents.
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20. Logging & Audit

- Audit every privileged action (review decisions, publishes, role changes, deletions/archives, settings) with actor, timestamp, before/after.
 - Operational logs for ingestion, OCR, indexing and AI jobs with correlation IDs; retained per policy.
 - Audit logs are immutable, queryable by admins, and exportable.
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21. Security Requirements

- Authentication with hashed credentials (bcrypt/argon2), email verification, secure password reset, optional SSO/MFA.
- Server-side RBAC on every route and action; least-privilege defaults; signed, expiring sessions.

- Input validation, output encoding, CSRF protection, rate limiting and abuse prevention on public endpoints (search, AI, upload, contact).
 - Encryption in transit and at rest; malware scanning of uploads; secrets in a managed vault; dependency and security scanning in CI.
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22. Backup & Disaster Recovery

- Automated daily backups of database and document store; periodic full + frequent incremental; encrypted, off-site copies.
 - Defined RPO $\leq 24\text{h}$ and RTO $\leq 8\text{h}$ for v1.0; restore procedures tested quarterly.
 - Point-in-time recovery for the database; versioned object storage for documents.
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23. Version Control Strategy

- **Code:** Git with trunk-based or GitFlow; protected main; PR review; CI gates (tests, lint, security); semantic versioning; tagged releases.
 - **Content:** document-level versioning with immutable history, amendment linkage and rollback; published-state changes audited.
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24. Future Expansion (v1.1+)

- Responsive/mobile refinement and native apps; dark mode.
 - Dedicated browsers for PRC, Research, Circulars, Notifications and Legal framework.
 - Engineering depth: PEC, NHA, PSDP, CSR, FIDIC plus specifications, drawings, manuals and testing standards (ASTM/BS/ISO).
 - International libraries (World Bank, ADB, IsDB); document comparison; saved searches & alerts; contributor recognition/leaderboards; public API.
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25. Acceptance Criteria

- All v1.0 functional requirements implemented and passing automated + manual test suites.
- RBAC enforced per the Permissions Matrix; no privileged action reachable by lower roles.

- Every AI answer shows official text separately, at least one citation, a confidence level and the disclaimer.
- No document publishable without source verification; full audit trail present.
- WCAG 2.1 AA verified for body text contrast, keyboard navigation and focus states; status conveyed by colour + text.
- Performance, availability and security NFR targets met in staging load/security tests.
- Responsive behaviour verified across mobile, tablet, laptop, desktop and ultra-wide.
- No dead controls: every button, link, filter, sort and menu produces a defined response or labelled placeholder.

26. Development Milestones

#	Milestone	Key deliverables
M0	Foundations	Architecture, environments, CI/CD, auth, RBAC, design-system components, responsive shell.
M1	Repository & Viewer	Document model, ingestion, repository browse, document detail + viewer.
M2	Search	Indexing, keyword + semantic + OCR search, facets, suggestions.
M3	AI Copilot	RAG pipeline, reading modes, citations, confidence, export.
M4	Contribution & Review	Upload, OCR, duplicate detection, moderation workflow, publishing.
M5	Accounts & Engagement	Bookmarks, collections, notifications, downloads, settings, analytics.
M6	Admin & Hardening	Admin dashboard, audit, backups, version control, security & performance hardening, UAT.

27. Risks

Risk	Impact	Mitigation
AI hallucination / mis-citation	High	Retrieval-grounding, mandatory citations, confidence levels, "no answer" path, disclaimers.
Inaccurate / unverified content	High	Mandatory source verification, versioning, correction channel.
OCR quality on poor scans	Medium	Confidence scoring, manual correction in review, re-OCR.
Scope creep beyond v1.0	Medium	Frozen scope, change-control, v1.1 backlog.
Copyright / attribution concerns	Medium	Public-source policy, clear attribution, takedown/correction process.
Performance at scale	Medium	Virtualisation, caching, async job queues, load testing.

28. Assumptions

- Documents ingested are publicly available and lawfully shareable.
- The frozen v1.0 design is the authoritative UI specification.
- An AI/LLM provider and OCR service are available and procured.
- Cloud hosting with managed database, object storage and search is available.

29. Dependencies

- LLM/embeddings provider (RAG + semantic search); OCR engine; search engine/vector store.
- Cloud platform (compute, managed DB, object storage, CDN); email/notification service; authentication/SSO provider.
- Frozen BPG design system and component library as the UI source of truth.

30. Final Release Scope (Version 1.0)

Ships in v1.0: Home, Search (keyword/semantic/OCR + facets + AI card), Repository Browser, Document Viewer (3 modes, find, panels), AI Copilot (5 reading modes, citations, confidence, export), Resources, Procurement Tools (9 calculators), About, Contact, Community Contribution, Contribution Review, Admin Dashboard, Authentication & RBAC, Bookmarks, Collections, Downloads, Notifications, Analytics, Settings — across responsive web with WCAG 2.1 AA body text, audit logging, backups and document/version control.

Deferred to v1.1+: dark mode, native mobile apps, dedicated PRC/Research/Circulars/Notifications/Legal browsers, expanded engineering & international libraries, document comparison, saved searches/alerts, public API.

Approval. This SRS reflects BPG Design Freeze v1.0 and is approved as the foundation for software architecture and development. Changes follow formal change-control against this baseline.